

ECE4132 Final Project: Task 4

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The following table (Table 1) of results shows the averages of each signal over 100 tests, average standard deviation across these tests, and the standard deviations of the means. π . As this system is run as an offline process **before** the night starts, we can run over 100,000 iterations for each test to get the most accurate and stable solutions, even though it is excessive. Note: π is the proposed price, n is the number of expected successful operators, and e is the error between n and **ref** (140)

	μ	μ of σ	σ of μ
π	\$11.56	\$2.26	\$0.00015
e	0.023	8.85	0.0078
n	139.97	8.85	0.0082

Table 1: System results after calibration

Whilst the deviation in mean over each test is remarkably small, the data in the table above shows that the error rate itself (< 140 drone operators) has quite a substantial standard deviation. This means that the estimated price of \$11.56 will result in an unpredictable output that could result in less than 140 drone operators. To be more exact, a mean of 140 operators implies that there will be 140 successful drone operators **half** of the time. The data shown in Figure 1 below better represents this fact and how unpredictable the system can be. As mentioned in Task 3, we are proposing a 2% increase to the \$11.56 amount to account for the inherent randomness in the Bernoulli trials, giving a new proposed price of \$11.79 per delivery. By re-computing the Bernoulli trials for this price, we find that the new mean is 148.9 drone operators, and a standard deviation of 8.825, meaning there is just under a 2.3% change the quota of drone operators will not be reached.

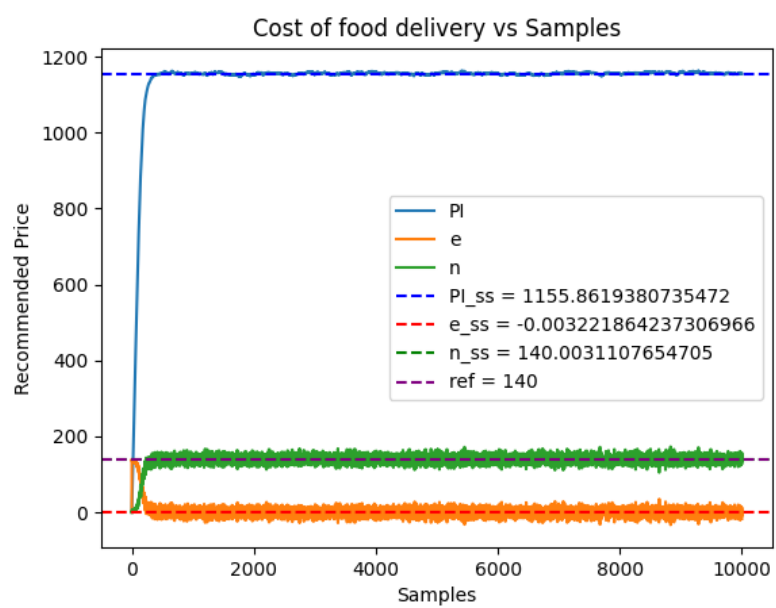


Figure 1: System results after calibration